Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 3 Quiz 1 - Forces and Dynamics

Mock Exam E · Form 1: Pulley Focus

Multiple Choice Questions: Select the best answer. (1 point each)

1. A ramp block has downhill weight component 8 N and maximum static friction 3 N. Vary its hanger weight W. Which hanger weights permit rest?

(A) Only W=8 N

(B) 5 N<=W<=11 N

(C) 0<=W<=3 N

(D) 8 N<=W<=11 N

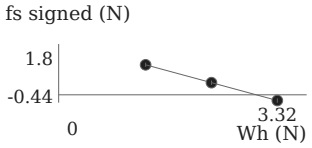
2. A stationary incline block with 3 N of downhill weight is connected to a hanger over an ideal pulley. Static friction is sufficient; count uphill friction positive. Double the block weight at fixed angle and test the same hanger weights. Rest remains possible. At the same X=2.32, what is the new minus old vertical reading?

(A) 3.21

(B) 2.58

(C) 3.63

(D) 3



3. A hanger rests while its connected incline block experiences uphill static friction. Which change first reduces that friction magnitude, keeping rest possible?

(A) Remove a little hanger mass

(B) Increase the ramp angle at fixed hanger weight

(C) Increase block weight at fixed tilt

(D) Add a little hanger mass

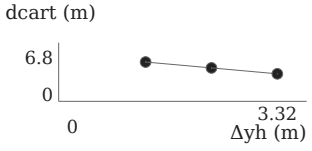
4. A cart and a hanging mass share a taut rope over one ideal fixed pulley. Measure downward hanger displacement and the cart distance from the pulley. Double the initial cart distance at the same hanger displacements. What is the new intercept minus the original intercept?

(A) 8.63

(B) 8

(C) 8.21

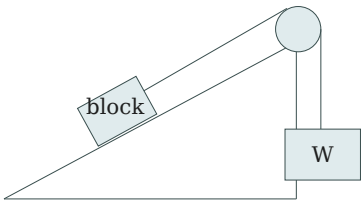
(D) 7.58



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. An incline block has downhill weight component 9 N and normal force 12 N. Static coefficient is 0.25 and hanger weight W is adjustable.

a. Mark friction at impending upward slip. **(1 pt)**



b. Find the full interval of W permitting rest and compare friction direction at its endpoints. **(3 pt)**

6. A horizontal cart slides toward an ideal pulley. Cart plus hanger mass is fixed at 5 kg; a constant 1 N resistance acts on the cart. Use g=10 m/s². Wh (N) -> a (m/s²): (1.32, 0.064); (2.32, 0.264); (3.32, 0.464).

a. Build the model for sliding cart and hanger from physics and predict both slope and intercept. **(2 pt)**

b. Double the resistance at fixed system mass. Predict the new slope and intercept and explain physically what changed. **(2 pt)**

7. A stationary block of mass M on an incline is connected to a hanger of mass m by an ideal rope. Static friction coefficient is μ. Derive both bounds on hanger mass for which rest is possible. **(3 pt)**